

Total No. of Questions : 4]

SEAT No. : 933

PD89

[Total No. of Pages : 2

[6410]-410

T.E. (Chemical Engineering) (Insem)
CHEMICAL REACTION ENGINEERING-I
(2019 Pattern) (Semester - II) (309348)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

UNIT-I

- Q1) a) In the 19th century, Dr. Henric Wilson collected the data of ozone depletion at Antarctica location. Based on the data, he came with conclusion that the rate of ozone depletion as below: [10]

$$-r_{O_3} = k [O_3]^2 [O_2]^{-1}$$

- i) What is the overall order of reaction?
 - ii) Suggest a two-step mechanism to explain this rate and state how you would further test this mechanism.
- b) Explain rate of reaction and give the various ways of expressing the reaction rate. [5]

OR

- Q2) a) Explain classification of chemical reaction. [7]
- b) Explain kinetic model for non-elementary reaction. [8]

P.T.O